6 The equation of a curve is

$y = x^4 - 8x^3 + 20$.

(a) Show that the gradient function, $\frac{dy}{dx}$, is

$\frac{dy}{dx} = 4x^2(x-6)$. **[2]**

(b) **(i)** On the axes in the Printed Answer Booklet, sketch the gradient function of y. **[2]**

(ii) State the coordinates of the intercepts of the gradient function of y. **[1]**

(c) Hence state the range of values of x for which $y = x^4 - 8x^3 + 20$ is an increasing function. Express your answer in set notation. **[1]**

